

AI Search Framework v2: Pairwise & Multi-Criterion Benchmark

Study: All-SLM Pipeline vs Multi-LLM Baseline Roster | **Evaluation:** N=80 Paired Trials | **Judge:** Anonymous Claude-3.5-Sonnet with Bradley-Terry Modeling

Executive Summary

The v2 AI Search Framework benchmark evaluates an all-SLM pipeline ($\leq 8B$) with a **bounded re-decomposition feedback loop** (max depth = 3) against 5 baseline models spanning parameter scales (8B to Frontier).

Core Analytical Discoveries:

- Pairwise Head-to-Head Strength:** All-SLM Pipeline achieves a **100.0% win rate vs Llama-8B**, **62.5% vs Qwen-32B**, and **25.0% vs Llama-70B** (winning 1 in 4 queries outright against 70B models).
- Criterion Attribution (Aggregation Bottleneck):** SLM Pipeline achieves near-parity on **Correctness (4.47 vs 4.64)** and **Completeness (4.64 vs 4.70)**, but incurs a **0.64-point Coherence penalty (4.11 vs 4.75)** due to judge bias toward single-voice prose. *The bottleneck is Aggregation, not Decomposition.*
- Compute Cost Savings:** Delivered **41.6% cost savings vs Llama-70B** (\$0.584 $\times$) and **88.9% savings vs Gemini-1.5-Pro** (\$0.111 $\times$).
- Feedback Loop Precision:** Re-decomposition reduced Graph Edit Distance (GED) from **3.57 $\rightarrow$ 1.50 (58.33% error reduction)**.

1. Pairwise Win Probability & Bradley-Terry Latent Elo Ratings

System Identifier	Param Scale	Win Rate vs SLM	SLM Win Rate vs Baseline	Bradley-Terry Elo	Tier Rank
Gemini-1.5-Pro	Frontier	100.0%	0.0%	1806.7	Tier 1 (Frontier API)
Qwen-2.5-72B	72.7B	100.0%	0.0%	1170.4	Tier 2 (Dense Open SOTA)
Llama-3.1-70B	70.6B	75.0%	25.0%	495.8	Tier 3 (Dense 70B)
All-SLM Pipeline v2	$\leq 8.0B$	—	—	227.5	Tier 4 (SLM Network)
Qwen-2.5-32B	32.5B	37.5%	62.5%	77.6	Tier 5 (Mid Monolithic)
Llama-3.1-8B	8.0B	0.0%	100.0%	-100.0	Tier 6 (Single Small Model)

2. Criterion-by-Criterion Performance Breakdown

Evaluating Correctness (40%), Completeness (35%), and Coherence (25%) on a 1.0–5.0 scale:

System	Mean Correctness	Mean Completeness	Mean Coherence	Composite Score	Coherence Gap vs 70B
All-SLM Pipeline v2	4.471	4.644	4.112	4.442	-0.638 (Voice Penalty)
Llama-3.1-8B	3.613	3.938	4.550	3.961	-0.200
Qwen-2.5-32B	4.344	4.450	4.500	4.420	-0.250
Llama-3.1-70B	4.637	4.700	4.750	4.688	0.000 (Reference)
Qwen-2.5-72B	4.806	4.800	4.750	4.790	0.000
Gemini-1.5-Pro	4.850	4.850	4.900	4.862	+0.150

Key Synthesis vs. Decomposition Diagnosis:

- On Single-Domain tasks (no multi-source stitching), SLM Pipeline Coherence is **4.75 / 5.0**.
- On 3+-Domain compound tasks, SLM Pipeline Coherence drops by **1.10 points to 3.65 / 5.0**, while Correctness remains strong at **4.55 / 5.0**.
- Actionable Insight:** Subtask decomposition and specialist SLMs are working properly; future architectural refinement must focus on single-voice Aggregator rewriting.

3. Complexity Tier & Domain Cluster Win-Rates

Stratum	N	SLM Pipeline Win-Rate	Llama-70B Win-Rate	Qwen-72B Win-Rate	Gemini-Pro Win-Rate
Single-Domain Control	20	20.00%	15.0%	25.0%	25.0%
2-Domain Compound	30	10.00%	16.7%	23.3%	16.7%
3+-Domain Compound	30	13.33%	13.3%	26.7%	16.7%
Code/Math Cluster	38	13.16%	15.8%	28.9%	18.4%
Cross-Domain Cluster	42	14.29%	14.3%	21.4%	19.0%

4. Cost Ratios & Feedback Loop Summary

- vs Llama-3.1-70B: **0.584x** cost ratio (41.6% savings, 95% CI [0.551, 0.616]).
- vs Gemini-1.5-Pro: **0.111x** cost ratio (88.9% savings, 95% CI [0.104, 0.117]).
- Feedback Loop GED:** Reduced from **3.57 → 1.50 (58.33% reduction in graph errors)**.

Reproducibility Citation: Traceable to results/v2_judge_deep_analysis.json and results/v2_eval_dev_master.jsonl.